IAT New Test 8

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

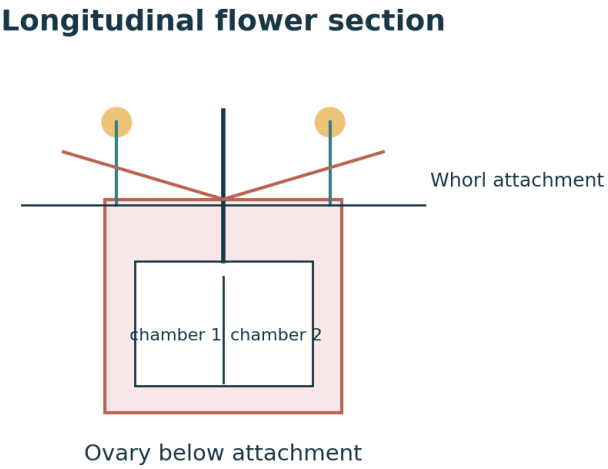
Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A moss sporophyte remains attached to a green gametophyte. When transfer of organic nutrients from the gametophyte is blocked, the sporophyte stops growing despite light and water. Which inference is best supported?
A) Its sporophyte depends nutritionally on the gametophyte
B) Its sporophyte is the only photosynthetic generation
C) Its gametophyte is always diploid
D) Spores cannot form by meiosis

Q2 The flower section in the figure places sepals, petals and stamens above an ovary with two chambers. Which floral description matches the attachment level?



A) Hypogynous with superior ovary
B) Epigynous with inferior ovary
C) Perigynous with a free superior ovary
D) A flower lacking a gynoecium

Q3 A membrane protein moves solute X inward even when inward X concentration exceeds the outside. A respiratory inhibitor stops the inward accumulation, but a separate dye confirms the membrane remains intact. Which mechanism is most consistent?
A) Simple diffusion through lipids
B) Facilitated diffusion down X’s gradient
C) Energy-coupled active transport
D) Osmotic movement of water

Q4 In matched leaves, a C_3 leaf fixes 120 units of carbon but loses 30 units to photorespiration; a C_4 leaf fixes 100 units and loses 5. Ignoring other losses, which has higher net carbon gain and by how much?
A) C_3 by 20
B) C_3 by 5
C) C_4 by 20
D) C_4 by 5

Q5 At a given tissue oxygen pressure, haemoglobin releases more O_2 after an increase in CO_2 and a fall in pH. Which shift of the dissociation curve explains this?
A) Rightward shift with lower affinity
B) Leftward shift with higher affinity
C) No shift because temperature is fixed
D) Complete loss of haemoglobin binding sites

Q6 An angiosperm's egg and each polar nucleus are haploid (n). After normal double fertilisation, what are the ploidies of the zygote and primary endosperm nucleus, respectively?
A) n and 2n
B) 2n and 3n
C) 3n and 2n
D) 2n and 2n

Q7 A dihybrid AaBb is test-crossed to aabb. Genes assort independently. Among offspring that show the dominant A phenotype, what fraction also shows dominant B?
A) 1/4
B) 1/3
C) 1/2
D) 3/4

Q8 A pathogen is cleared from immunised mice. Transferring purified serum to naive mice fails to protect, but transferring antigen-experienced T lymphocytes protects; heat-treated T cells are a negative control. Which defence is most directly implicated?
A) Only circulating antibody
B) Innate complement alone
C) Red-blood-cell immunity
D) Cell-mediated adaptive immunity

Q9 A PCR forward primer binds the bottom template just left of a target region. Which orientation must a reverse primer have on the top strand to amplify the bounded segment over cycles?
A) Its 3′ end points toward the forward primer and target interior
B) Its 5′ end points toward the target interior
C) Both primers bind the same strand facing right
D) A reverse primer is unnecessary after the first cycle

Q10 A producer population captures 900 kJ m^{-2} yr^{-1} as gross primary productivity and uses 350 kJ m^{-2} yr^{-1} in its own respiration. What is its net primary productivity?
A) 350
B) 550
C) 900
D) 1250

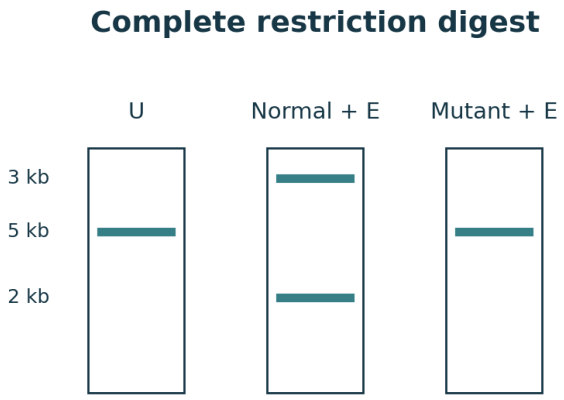
Q11 A woman is heterozygous for an X-linked recessive allele and her partner is affected. Given that their daughter is unaffected, what is the probability she is a carrier?
A) 0
B) 1/2
C) 1
D) 2/3

Q12 A two-gene test cross yields parental classes 420 and 380, and recombinant classes 110 and 90. What is the recombination fraction and which conclusion is warranted?
A) 10%; complete linkage
B) 20%; independent assortment
C) 50%; independent assortment
D) 20%; linkage with recombination

Q13 At a nephron, filtration rate is 100 mL min^{-1} and plasma concentration of substance X is 2 mg mL^{-1}. Urinary excretion is 50 mg min^{-1}. Assuming no secretion, what is net tubular reabsorption?
A) 150 mg min^{-1}
B) 50 mg min^{-1}
C) 200 mg min^{-1}
D) 250 mg min^{-1}

Q14 Two cell fractions both contain membrane-bound vesicles. Fraction P has maximal catalase activity; Q has acid phosphatase strongest at acidic pH. Which assignment uses both enzyme controls?
A) P lysosome; Q peroxisome
B) P peroxisome; Q lysosome
C) Both mitochondria
D) P and Q are necessarily identical organelles

Q15 The gel figure shows a 5 kb DNA molecule uncut in lane U. After restriction enzyme E, the normal molecule gives 3 kb and 2 kb bands; a mutant sample gives only a 5 kb band. Assuming complete digestion in both samples, what changed?



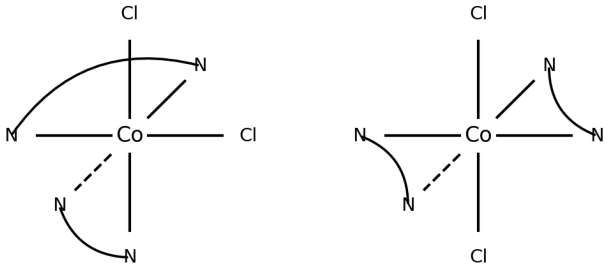
A) An extra E recognition site appeared
B) The mutant molecule shrank to 2 kb
C) The E site was lost or altered in the mutant
D) The enzyme was absent from the mutant reaction

Chemistry

Q16 A molecule has one carbon triple-bonded to nitrogen and single-bonded to carbon, as in R–C≡N. How many σ and π bonds are in just the C≡N unit, and what is hybridisation of that carbon?
A) 3σ, 0π; sp^3
B) 2σ, 1π; sp^2
C) 1σ, 2π; sp^2
D) 1σ, 2π; sp

Q17 The diagrams show the two geometrical arrangements of octahedral $[Co(en)_2Cl_2]^+$, with each en joining two adjacent donor sites. Counting mirror-image forms separately but ignoring conformers, how many configurational stereoisomers exist in total, and how many of them are optically active?

Octahedral positions and connected donor pairs



Arrangement X Arrangement Y

Each joined N pair belongs to one en ligand; projections are schematic

A) 3 total; 2 optically active
B) 2 total; 0 optically active
C) 4 total; 4 optically active
D) 3 total; 3 optically active

Q18 For the nitrate ion, a proposed Lewis contributor has N double-bonded to one O and single-bonded to two O atoms, with octets satisfied. What formal charge does nitrogen have?
A) 0
B) +1
C) −1
D) +2

Q19 An atom has three electrons in a set of three degenerate p orbitals. Which lowest-energy occupancy follows Hund’s rule before pairing?
A) One orbital doubly occupied, one singly, one empty
B) All three paired somehow
C) One electron in each orbital with parallel spins
D) Three electrons in one orbital

Q20 Which carbocation can delocalise its positive charge into an adjacent benzene π system, unlike an ordinary primary alkyl carbocation?
A) $CH_3CH_2^+$
B) $(CH_3)_3C^+$
C) $CH_3CH_2CH_2^+$
D) $C_6H_5CH_2^+$

Q21 Equimolar ethanal and propanone compete for a limited amount of $NaBH_4$ under identical conditions. Early analysis shows faster disappearance of ethanal. Which structural reason best supports this trend?
A) The aldehyde carbonyl is less hindered and less alkyl-donated
B) The ketone has no carbonyl carbon
C) $NaBH_4$ oxidises ethanal exclusively
D) Ethanal cannot be reduced at all

Q22 An aryl diazonium chloride is prepared at 0–5 °C. One aliquot couples with alkaline phenol to give an azo dye; a second is warmed with water and releases N_2. Which product is expected from the second aliquot, and why keep the first cold?
A) Aniline; cold prevents reduction
B) Phenol; cold retains diazonium long enough for coupling
C) Benzene; cold accelerates hydrolysis
D) Chlorobenzene; cold supplies chloride substitution

Q23 Complete hydrolysis of a purified tripeptide Gly–Ala–Gly yields which amino-acid counts?
A) Three glycine, no alanine
B) One glycine, two alanine
C) Two glycine, one alanine
D) One of each plus a dipeptide

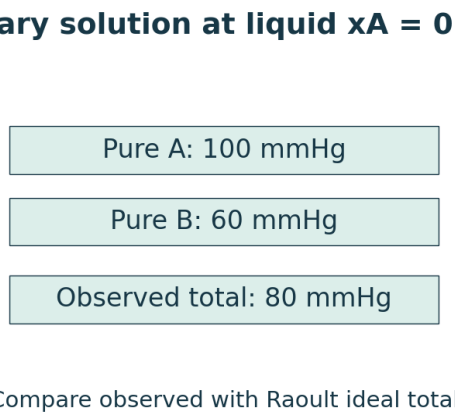
Q24 A reaction's half-life is 20 min when initial concentration is 0.10 M and remains 20 min when it is 0.20 M. Which order is most consistent under identical conditions?
A) Zero
B) Second
C) Third
D) First

Q25 A saturated AgCl solution is in contact with solid AgCl and a reservoir that holds free NH_3 at 1.0 M. Use $K_{sp} = 10^{-10}$ and $[Ag(NH_3)_2^+]/([Ag^+][NH_3]^2) = 10^8$. No chloride is initially added, and free Ag^+ is negligible compared with its complex. What is the approximate total dissolved silver concentration?
A) 0.10 M
B) 10^{-5} M
C) 10^{-10} M
D) 1.0 M

Q26 Acidifying yellow chromate solution gives orange dichromate through $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$. What happens to chromium oxidation number?
A) +6 to +3
B) It stays +6
C) +6 to +7
D) +3 to +6

Q27 A reaction has $\Delta H = -40$ kJ mol^{-1} and $\Delta S = -100$ J mol^{-1} K^{-1}, approximately constant. At what temperatures is it spontaneous at constant pressure?
A) Only T>400 K
B) At all T
C) Only T<400 K
D) At no T

Q28 From the vapour-pressure figure at the marked liquid composition, compare the measured total with the Raoult-law prediction. Which deviation and intermolecular interpretation fit?



A) Negative deviation, stronger A–B attraction
B) Ideal behaviour at every composition
C) No conclusion because xB is unknown
D) Positive deviation, weaker A–B attraction

Q29 Reductive ozonolysis of a cyclic alkene produces one open-chain molecule with an aldehyde at each end and four methylene groups between them. What starting alkene is consistent?
A) Cyclohexene
B) Hex-1-ene
C) Cyclopentene
D) Benzene

Q30 Alcoholic KOH eliminates HBr from 2-bromobutane. Ignoring minor stereochemical products, which alkene is generally the major Zaitsev product?
A) But-1-ene
B) But-2-ene
C) Butane
D) 2-methylpropene

Mathematics

Q31 The hyperbola $x^2/9 - y^2/16 = 1$ has its transverse semiaxis doubled while its conjugate semiaxis stays unchanged. What are the new asymptote slopes?
A) ±4/3
B) ±3/2
C) ±2/3
D) ±3/4

Q32 For $f(x) = x^3 - 6x^2 + 9x$, on which intervals is f strictly increasing?
A) (1,3) only
B) All real x
C) $(-\infty, 3)$ only
D) $(-\infty, 1)$ and $(3, \infty)$

Q33 Using $u = 1 + x^2$ and transformed integration bounds, what is the exact value of $\int_0^1 x\sqrt{1+x^2}\,dx$?
A) $(2\sqrt{2} - 1)/3$
B) $(2\sqrt{2} + 1)/3$
C) $2\sqrt{2} - 1$
D) $\sqrt{2} - 1)/3$

Q34 On S={1,2,3,4,5,6}, R contains all diagonal pairs and exactly the additional pairs (1,2),(2,1),(2,3),(3,2),(4,5),(5,4). What is the minimum number of ordered pairs that must be added to make R an equivalence relation with exactly two equivalence classes?
A) 2
B) 6
C) 8
D) 14

Q35 For $A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 1 \\ 13 & 4 \end{pmatrix}$, AX=B. What is the trace of X?
A) −2
B) 1
C) 0
D) 2

Q36 A fair coin is tossed three independent times. What is the probability of at least one head?
A) 1/8
B) 3/8
C) 3/4
D) 7/8

Q37 How many integers from 1 through 100 are divisible by 2 or 3 but not by 5?
A) 54
B) 67
C) 40
D) 60

Q38 What is the term independent of x in $(x^2 + x^{-1})^9$?
A) 36
B) 84
C) 126
D) 252

Q39 For which a does the infinite series $\sum_{n=0}^{\infty}(a-1)^n$ converge and have sum 3?
A) 1/3
B) 4/3
C) 5/3
D) 2

Q40 Define f(x)=(x²−a²)/(x−a) for x≠a and f(a)=b. If f is continuous at a and b=6, find a and the derivative at a.
A) a=6, f′=1
B) a=3, f′=2
C) a=6, f′=2
D) a=3, f′=1

Q41 Let y solve y′+y/x=2x for x>0 with y(1)=1. Find $\int_1^2 y(x)/x\,dx$.
A) 7/6
B) 9/5
C) 17/6
D) 3/2

Q42 The planes x+y+z=3 and x−y+z=1 meet along a line. Which parametrisation describes their intersection?
A) (t,0,3−t)
B) (t,1,2−t)
C) (t,2,1−t)
D) (t,1,3−t)

Q43 For a=(1,0,1) and b=(0,2,0), which unit vector along a×b has positive z component?
A) (1/√2,0,−1/√2)
B) (1,0,1)
C) (−1/√2,0,1/√2)
D) (−1,0,1)

Q44 Random variable X has mean 3 and variance 4. Y=aX+b has mean −1 and variance 36, with a<0. What are a and b?
A) a=3,b=−10
B) a=−3,b=−8
C) a=3,b=8
D) a=−3,b=8

Q45 How many solutions of tan x=sin(2x) are in the closed interval [0,2π]?
A) 7
B) 5
C) 3
D) 6

Physics

Q46 From the same origin at t=0, particle A moves at constant +6 m s^{-1}; particle B starts at rest with constant +2 m s^{-2} acceleration. When do they next meet?
A) 3 s
B) 6 s
C) 9 s
D) 12 s

Q47 A velocity-time graph rises linearly from 0 to 72 km h^{-1} in 10 s. What displacement is represented by its triangular area?
A) 50 m
B) 72 m
C) 100 m
D) 200 m

Q48 A water-filled vessel accelerates upward at 2 m s^{-2}. What is gauge pressure at depth 0.50 m below its free surface? Use ρ=1000 kg m^{-3}, g=10 m s^{-2}.
A) 4000 Pa
B) 5000 Pa
C) 5500 Pa
D) 6000 Pa

Q49 A uniform disc of original mass M and radius R has a concentric hole of radius R/2 removed. Express remaining body's moment of inertia about the central perpendicular axis in terms of its remaining mass m and R.
A) (5/8)mR^2
B) (1/2)mR^2
C) (3/4)mR^2
D) (15/32)mR^2

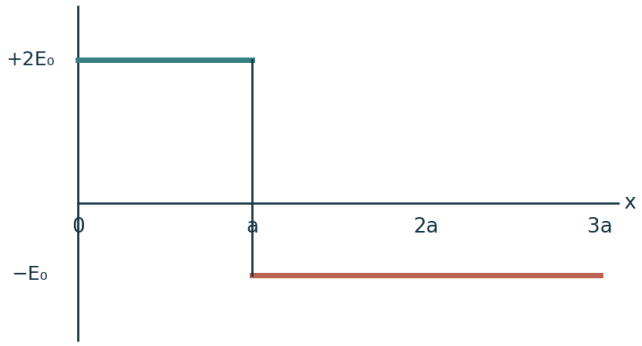
Q50 A heat engine completes a cycle, absorbing 500 J from the hot reservoir and rejecting 300 J to the cold reservoir. Find work per cycle and thermal efficiency.
A) 300 J, 60%
B) 200 J, 40%
C) 500 J, 40%
D) 200 J, 60%

Q51 Two identical springs of constant k support the same mass m. What is the ratio of oscillation period with springs in series to period with springs in parallel?
A) 1/2
B) 1
C) 2
D) 4

Q52 Two coherent waves arrive with geometric path difference λ/2, while source 2 has an initial phase lead of π relative to source 1. What is their interference at that point?
A) Complete destructive for equal amplitudes
B) No stable pattern
C) The waves have unequal frequencies
D) Constructive for equal amplitudes

Q53 From the signed electric-field graph between 0 and 3a, what are V(2a)−V(0) and V(3a)−V(0), respectively?

Signed electric field E_x versus x



A) $(-E_0a, 0)$
B) $(+E_0a, 0)$
C) $(0, -E_0a)$
D) $(-2E_0a, +E_0a)$

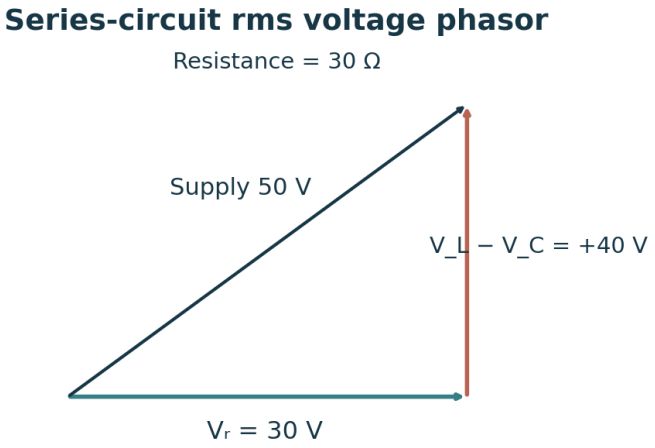
Q54 A 12 V battery with internal resistance 1 Ω powers an external 5 Ω resistor. What are current and terminal voltage while discharging?
A) 2 A, 12 V
B) 2 A, 10 V
C) 2.4 A, 10 V
D) 12 A, 2 V

Q55 Initially 800 radioactive A nuclei decay to stable B with half-life 2 h. After 4 h all accumulated B is removed; A remains. How many new B nuclei form in the next 2 h?
A) 50
B) 200
C) 100
D) 400

Q56 An ideal inductor L=0.50 H carries 2 A, then its current rises to 4 A. How much extra magnetic energy is stored?
A) 1 J
B) 2 J
C) 4 J
D) 3 J

Q57 A prism has apex angle 60° and refractive index √3 in air. At minimum deviation, what is the deviation angle?
A) 60°
B) 30°
C) 45°
D) 90°

Q58 From the rms phasor diagram for a series circuit, determine the current and whether it leads or lags the supply.



A) 5/3 A, leading
B) 1 A, lagging
C) 1 A, leading
D) 5/3 A, lagging

Q59 A sinusoidal 50 Hz supply feeds ideal half-wave and full-wave rectifiers. What are their output ripple repetition frequencies, half-wave then full-wave?
A) 100 Hz, 50 Hz
B) 50 Hz, 50 Hz
C) 50 Hz, 100 Hz
D) 100 Hz, 100 Hz

Q60 At normal incidence a film of index 1.5 lies on glass of index 1.4. For reflected light λ=600 nm in air, what is the least positive film thickness for constructive reflection? Assume no absorption.
A) 50 nm
B) 200 nm
C) 150 nm
D) 100 nm

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A bryophyte sporophyte dependence

Answer: (A) Nutritional dependence on the gametophyte.

Why: The intervention leaves external light and water available but blocks the transfer route from the attached gametophyte. Arrested growth therefore supports dependence on its parent tissue for at least some resources. It does not prove complete absence of sporophyte photosynthesis; the comparison isolates nutrient support.

Why not the others: (B) overclaims exclusive photosynthesis; (C) confuses the haploid gametophyte with diploid sporophyte; (D) concerns a later spore-production mechanism not tested here.

Key point: Use the controlled change to identify dependence, not every metabolic capacity.

Q2 / B flower whorl position

Answer: (B) Epigynous with inferior ovary.

Why: The receptacle encloses the ovary below the level where the other floral whorls appear attached. Their position, not the number of chambers, defines an epigynous flower and an inferior ovary. The two chambers confirm an ovary is present but do not independently decide superiority.

Why not the others: (A) requires whorls below the ovary; (C) places the ovary free within a cup rather than enclosed below attachment; (D) contradicts the pictured ovary.

Key point: Compare whorl attachment with ovary position.

Q3 / C membrane transport control

Answer: (C) Energy-coupled active transport.

Why: Accumulation against X's concentration gradient needs an energy source or coupling to another downhill gradient. The inhibitor removes the inward movement while membrane integrity persists, supporting an energy-dependent carrier rather than leakage or destruction of the barrier. The second dye control is needed to distinguish transport loss from membrane rupture.

Why not the others: (A) and (B) cannot drive X uphill by its own gradient; (D) concerns water rather than X accumulation.

Key point: Combine gradient direction with an energy and integrity control.

Q4 / D photorespiration contrast

Answer: (D) C_4 by 5 units.

Given: Net gain equals stated fixation minus photorespiratory loss.

Formula/Principle: Compare $120-30$ with $100-5$.

Step 1: C_3 net=90 units.

Step 2: C_4 net=95 units, exceeding C_3 by 5.

Why: The C_4 carbon-concentrating mechanism reduces oxygenation loss even when its gross fixation in this particular comparison is lower. Net rather than gross carbon is what the question asks; the stated simplified account excludes other respiration terms.

Why not the others: (A) compares gross values only; (B) assigns the five-unit difference to the wrong leaf; (C) confuses gross difference with net advantage.

Key point: Subtract the appropriate loss before ranking pathways.

Q5 / A gas transport and dissociation

Answer: (A) Rightward shift and reduced affinity.

Why: Higher CO_2 and lower pH promote the Bohr effect, stabilising the lower-affinity state of haemoglobin. At the same tissue oxygen pressure, lower saturation means more O_2 has been unloaded. Temperature need not change for this response.

Why not the others: (B) would hinder unloading; (C) ignores pH and CO_2; (D) invents destruction of binding sites.

Key point: Relate a curve shift to saturation at the same pressure.

Q6 / B double-fertilisation ploidy

Answer: (B) 2n and 3n.

Given: Two haploid male gametes; egg n and two polar nuclei n+n.

Formula/Principle: One sperm fuses with egg; the other joins the two polar nuclei.

Step 1: Zygote n+n=2n.

Step 2: Primary endosperm nucleus n+n+n=3n.

Why: These are separate fusion events in the same embryo sac. Counting the polar nuclei as a single n nucleus before fusion would understate endosperm ploidy; the prompt specifies both are haploid.

Why not the others: (A) omits one gamete contribution to each; (C) reverses product roles; (D) ignores the second polar nucleus.

Key point: Track each participating haploid nucleus.

Q7 / C Mendelian two-locus segregation

Answer: (C) One half.

Given: Test-cross offspring are AaBb, Aabb, aaBb, aabb equally often.

Formula/Principle: Condition on the two A-dominant classes.

Step 1: A-dominant probability is 2/4.

Step 2: Joint A- and B-dominant probability is 1/4, so conditional fraction (1/4)/(2/4)=1/2.

Why: Independent segregation makes all four gamete classes equally likely, but the observed A condition removes half the sample space. This is distinct from the unconditional probability of both dominant traits.

Why not the others: (A) is unconditional joint probability; (B) uses an incorrect denominator; (D) counts an A-only class as B-dominant.

Key point: Condition on the phenotype specified in the question.

Q8 / D antibody versus cellular defence

Answer: (D) Cell-mediated adaptive immunity.

Why: Protective activity follows viable antigen-experienced T cells rather than the transferred antibody-containing serum. The heat-treated control makes nonspecific material carried with the cells less plausible. This supports a cellular effector or helper mechanism; it does not claim antibodies never contribute in other infections.

Why not the others: (A) predicts serum protection; (B) does not explain the T-cell-specific transfer result; (C) assigns immune specificity to erythrocytes.

Key point: Use transfer and viability controls to locate the protective arm.

Q9 / A PCR primer orientation

Answer: (A) The reverse primer's 3′ end faces inward.

Why: Polymerase extends only from primer 3′ ends, so the two primers must face each other across the segment on opposite templates. Outward-facing reverse extension would copy away from the bounded target and not yield exponential accumulation of that product.

Why not the others: (B) confuses 5′ orientation with extension direction; (C) loses opposite-strand bounding; (D) ignores recurring priming each cycle.

Key point: Draw both 3′ ends pointing into the amplicon.

Q10 / B primary versus secondary productivity

Answer: (B) 550 kJ m^{-2} yr^{-1}.

Given: Gross fixation 900, producer respiration 350 in the same units.

Formula/Principle: NPP=GPP−producer respiration.

Step 1: Subtract 350 from 900.

Step 2: Net stored production=550 kJ m^{-2} yr^{-1}.

Why: Net primary productivity is the organic energy left after producer metabolism and available for growth and consumers. Consumer respiration has not been supplied and must not be subtracted a second time.

Why not the others: (A) is respiration; (C) is gross production; (D) adds rather than subtracts respiratory use.

Key point: Distinguish gross fixation from net producer storage.

Q11 / C conditional pedigree risk

Answer: (C) Certain (probability 1).

Given: Father X^aY gives X^a to every daughter; mother gives X^A or X^a equally.

Formula/Principle: Daughters are X^AX^a (unaffected carrier) or X^aX^a (affected).

Step 1: Unaffected daughter must have received maternal X^A.

Step 2: She still inherits paternal X^a and is therefore a carrier.

Why: The observed unaffected phenotype removes the affected homozygous daughter from the conditional set, unlike a generic carrier×carrier autosomal calculation. Father's affected X chromosome cannot be absent from a biological daughter in this simple model.

Why not the others: (A) ignores paternal X^a; (B) is the unconditional carrier fraction among daughters; (D) imports an autosomal unaffected-sibling ratio.

Key point: Condition after tracing the sex-specific paternal chromosome.

Q12 / D linkage test-cross interpretation

Answer: (D) 20%; linkage with recombination.

Given: Total 1000 offspring; recombinant total 200.

Formula/Principle: Recombination fraction = recombinant/total.

Step 1: (110+90)/1000=0.20.

Step 2: Frequency below 0.50 and observed recombinants support linkage with crossing over.

Why: Parental classes dominate, indicating the loci do not assort as independent 1:1:1:1 classes in this cross. Recombinant offspring nonetheless show that exchange or equivalent recombination occurred; a 20% map fraction is not complete linkage.

Why not the others: (A) halves the recombinant count; (B) assigns 20% to independent assortment; (C) treats the unequal parental/recombinant classes as the 50% independent-assortment limit.

Key point: Count both recombinant classes and interpret below-50% fraction.

Q13 / A nephron clearance and reabsorption

Answer: (A) 150 mg min^{-1}.

Given: Filtered load is $100 \times 2 = 200$ mg min^{-1}.

Formula/Principle: Excretion=filtration−reabsorption+secretion.

Step 1: Secretion is stipulated zero.

Step 2: Reabsorption$=200-50=150$ mg min^{-1}.

Why: A clearance measurement alone does not identify transport direction unless secretion is constrained. Here the no-secretion condition allows the deficit from filtered to excreted load to be assigned to reabsorption.

Why not the others: (B) reports excretion; (C) assumes complete reabsorption; (D) adds excretion to filtration.

Key point: Write a mass balance with all tubular fluxes.

Q14 / B organelle fractionation

Answer: (B) P peroxisome; Q lysosome.

Why: Peroxisomes contain catalase to break down hydrogen peroxide. Lysosomes carry acid hydrolases, including acid phosphatase that functions best in an acidic compartment. Vesicle appearance alone cannot discriminate these fractions; the two enzyme readouts do.

Why not the others: (A) reverses enzyme markers; (C) lacks the paired diagnostic activities; (D) ignores the distinct biochemical profiles.

Key point: Identify organelles from functional markers plus structural evidence.

Q15 / C gel fragment interpretation

Answer: (C) The E recognition site was lost or altered.

Why: Normal digestion divides the 5 kb molecule into fragments whose lengths sum to 5 kb. A mutant that remains 5 kb under the same complete-digestion control no longer presents the cuttable site. The control rules out simply blaming an inactive enzyme in that lane.

Why not the others: (A) would add smaller fragments; (B) contradicts the 5 kb band; (D) contradicts the stipulated complete digestion control.

Key point: Compare fragment sum and the controlled enzyme condition.

Chemistry

Q16 / D hybridisation and sigma-pi count

Answer: (D) One sigma, two pi; nitrile carbon sp.

Why: A triple bond consists of one head-on sigma overlap and two side-on pi overlaps. The nitrile carbon has two regions of electron density, one to R and one to N, giving linear sp hybridisation. Count only the C≡N unit for the requested bonds; the R–C sigma bond is separate.

Why not the others: (A) counts all bonds as sigma and wrong geometry; (B) describes a double bond; (C) has the right bond partition but wrong sp^2 hybridisation.

Key point: Separate bond multiplicity from the atom's steric number.

Q17 / A Geometrical and optical multiplicity of a chelated octahedron

Answer: (A) Three total, of which two are optically active.

Given: The two chloride sites can be adjacent or opposite; en is bidentate and its donor pairs remain connected as shown.

Formula/Principle: Count geometrical arrangements first, then distinguish non-superposable mirror partners within each.

Step 1: Adjacent chlorides define the cis arrangement; opposite chlorides define trans.

Step 2: The cis arrangement has two non-superposable mirror forms because of the connected chelate rings.

Step 3: The trans arrangement has a symmetry plane and contributes one achiral form.

Step 4: Total $= 2 + 1 = 3$; optically active forms $= 2$.

Why: A ligand count or high/low-spin comparison cannot decide this stereochemical count. The chelate connections in the diagram must be preserved when testing whether a mirror image can be rotated onto the original.

Why not the others: (B) counts geometries but merges the cis mirror pair; (C) incorrectly splits trans into an optical pair; (D) treats the trans form as chiral.

Key point: Count geometry and optical multiplicity in separate stages.

Q18 / B formal-charge minimisation

Answer: (B) +1.

Given: Nitrogen has five valence electrons, four shared bond pairs and no lone pair in this contributor.

Formula/Principle: Formal charge = valence − nonbonding − half bonding electrons.

Step 1: Four bond pairs place eight bonding electrons around N.

Step 2: FC(N)=5−0−8/2=+1.

Worked derivation:

$$\mathrm{FC(N)} = 5 - N_{\text{nonbonding}} - \tfrac{1}{2}N_{\text{bonding}} = 5 - 0 - \tfrac{8}{2} = +1.$$

Why: The two singly bonded oxygens each carry −1, giving net −1 for nitrate when nitrogen is +1. Formal charge is bookkeeping within a contributor, not a statement that nitrogen is a free N^+ ion.

Why not the others: (A) omits the charge needed for net −1; (C) reverses sign; (D) overcounts the bond electron assignment.

Key point: Check atom charges against the whole ion's charge.

Q19 / C orbital occupancy and energy

Answer: (C) One parallel-spin electron in each p orbital.

Why: Filling degenerate orbitals singly with parallel spins maximises exchange stabilisation before pairing costs energy. The three p orbitals are distinct spatial states; Pauli's rule also forbids three electrons in a single orbital. The energy comparison assumes no external splitting that destroys degeneracy.

Why not the others: (A) pairs prematurely; (B) requires six electrons for three pairs; (D) violates the two-electron orbital limit.

Key point: Apply Hund before pairing in equal-energy orbitals.

Q20 / D stability of reactive intermediates

Answer: (D) Benzyl cation.

Why: The positively charged benzylic carbon has an empty p orbital adjacent to the aromatic π system; resonance can distribute the charge over the ring. Hyperconjugation can stabilise a tertiary alkyl cation, but it is not the specified aromatic delocalisation.

Why not the others: (A) is ordinary primary alkyl; (B) is tertiary without an adjacent benzene ring; (C) is another primary alkyl cation lacking conjugation.

Key point: Identify the orbital that can overlap with a neighbouring π system.

Q21 / A carbonyl addition selectivity

Answer: (A) Aldehyde is generally more accessible/electrophilic.

Why: Ethanal has one alkyl substituent and a hydrogen at its carbonyl carbon; propanone has two alkyl groups, increasing steric hindrance and electron donation. Hydride addition can occur to both, but the stated early-rate difference is consistent with a more reactive aldehyde under comparable conditions.

Why not the others: (B) contradicts propanone structure; (C) reverses reducing-agent action; (D) contradicts observed disappearance.

Key point: Compare both steric and electronic effects on carbonyl addition.

Q22 / B diazonium cold-warm response

Answer: (B) Phenol; cold permits diazonium coupling before hydrolysis.

Why: Diazonium ions are more persistent at low temperature and undergo electrophilic azo coupling with activated phenol. Warming in water promotes replacement of the diazonium group by OH with loss of N_2, furnishing phenol. The two aliquots distinguish competition between coupling and hydrolysis rather than a change of starting aryl skeleton.

Why not the others: (A) predicts reduction absent here; (C) omits hydrolytic OH incorporation; (D) assumes a copper-mediated chloride substitution not supplied.

Key point: Temperature and reagents determine which diazonium pathway dominates.

Q23 / C peptide hydrolysis products

Answer: (C) Two glycine and one alanine.

Why: Hydrolysis cleaves both peptide bonds but does not change residue identities. The sequence names three residues, with glycine at both ends; complete rather than partial cleavage yields only free amino acids. A residual dipeptide would indicate incomplete reaction.

Why not the others: (A) changes alanine into glycine; (B) reverses multiplicities; (D) retains a bond despite complete hydrolysis.

Key point: Count residues before and after bond cleavage.

Q24 / D integrated half-life relation

Answer: (D) First order.

Given: Half-life independent of doubled initial concentration.

Formula/Principle: For first order $t_{1/2} = \ln 2/k$; for zero and second order it depends on initial concentration.

Step 1: The measured two half-lives are equal.

Step 2: This matches concentration-independent first-order half-life.

Worked derivation:

$$t_{1/2} = \ln 2/k \quad \text{for first order;} \qquad \frac{t_{1/2}(0.20\ \mathrm{M})}{t_{1/2}(0.10\ \mathrm{M})} = \frac{20}{20} = 1.$$

Why: The comparison changes only starting concentration, so it diagnoses the functional dependence of half-life rather than the rate's numerical value. Two measurements support the first-order model but do not establish an exact mechanism.

Why not the others: (A) predicts half-life doubles with concentration; (B) predicts it halves; (C) has another concentration dependence.

Key point: Use controlled concentration change to test kinetic order.

Q25 / A Dissolution coupled to supplied complex-formation equilibrium

Answer: (A) Approximately 0.10 M.

Given: Each dissolved AgCl supplies one chloride; the ammonia reservoir fixes its free concentration despite complex formation.

Formula/Principle: Combine dissolution and complex formation, retaining free-ion concentrations in K_{sp}.

Step 1: Let total dissolved silver be s. Then $[\mathrm{Cl}^-] = s$ and $[\mathrm{Ag(NH_3)_2}^+] \simeq s$.

Step 2: The formation relation gives $[\mathrm{Ag}^+] \simeq s/10^8$.

Step 3:

$$K_{sp} \simeq \frac{s}{10^8} s = 10^{-10} \Rightarrow s^2 = 10^{-2} \Rightarrow s = 0.10 \text{ M}.$$

Step 4: Free silver is about 10^{-9} M, confirming that it is negligible relative to 0.10 M complex.

Why: Complex formation removes free silver ions without removing total dissolved silver, so more solid can dissolve while the free-ion product remains fixed. The reservoir assumption is essential; the free ammonia concentration could not otherwise be treated as unchanged at this solubility.

Why not the others: (B) gives pure-water solubility with no complexing ligand; (C) treats the solubility product as a concentration; (D) omits the square root in the combined equilibrium.

Key point: Distinguish free-ion concentration from total analytical solubility.

Q26 / B oxidation-number constraints

Answer: (B) It remains +6.

Why: In chromate, x+4(−2)=−2 yields x=+6. In dichromate, 2x+7(−2)=−2 yields x=+6 again. The colour shift is an acid-base equilibrium involving oxygen/protons, not an electron-transfer redox event.

Why not the others: (A) invents reduction; (C) invents oxidation; (D) assigns a +3 starting state that chromate does not have.

Key point: Calculate both oxidation states before calling a colour change redox.

Q27 / C Gibbs-temperature boundary

Answer: (C) T<400 K.

Given: Convert entropy to −0.100 kJ mol⁻¹ K⁻¹.

Formula/Principle: ΔG=ΔH−TΔS=−40+0.100T kJ mol⁻¹.

Step 1: Set ΔG<0.

Step 2: 0.100T<40, so T<400 K; at 400 K equilibrium boundary.

Worked derivation:

$$\Delta G = -40 - T(-0.100) = -40 + 0.100T < 0 \Rightarrow 0 < T < 400 \text{ K}.$$

Why: Negative enthalpy favours the reaction, but negative entropy penalises it increasingly at high temperature. Unit conversion is essential before equating the two terms. The boundary is not included in a strictly spontaneous range.

Why not the others: (A) reverses the temperature effect; (B) ignores the high-temperature entropy penalty; (D) ignores low-temperature favourability.

Key point: Convert units and solve strict ΔG inequality.

Q28 / D partial-vapour-pressure table

Answer: (D) Positive deviation, weaker unlike-molecule attraction.

Given: xB=0.75; ideal total 0.25(100)+0.75(60)=70 mmHg.

Formula/Principle: Compare observed 80 with ideal 70.

Step 1: Observed exceeds ideal by 10 mmHg.

Step 2: Positive deviation is consistent with unlike A–B interactions weaker than like interactions.

Worked derivation:

$$p_{\text{ideal}} = 0.25(100) + 0.75(60) = 70 \text{ mmHg}, \quad p_{\text{observed}} - p_{\text{ideal}} = 80 - 70 = +10 \text{ mmHg}.$$

Why: Weaker unlike attraction allows molecules to escape more readily, elevating vapour pressure. A single composition establishes deviation at that composition, not the shape of the entire phase diagram. The liquid fractions still sum to one.

Why not the others: (A) predicts pressure below 70; (B) contradicts the measured 80; (C) ignores that xB=1−xA=0.75.

Key point: Compute ideal pressure first, then infer sign of deviation.

Q29 / A ozonolysis reconstruction

Answer: (A) Cyclohexene.

Why: Cutting the C=C in a six-membered ring opens it into a six-carbon chain with aldehydes at both former alkene carbons: OHC–$(CH_2)_4$–CHO. A terminal acyclic alkene would produce two separate carbonyl molecules instead of one dialdehyde. The carbon count and single-molecule outcome both matter.

Why not the others: (B) gives separate fragments; (C) yields three intervening methylenes; (D) does not undergo this simple isolated-alkene cleavage.

Key point: Reconnect carbonyl carbons to reconstruct a cleaved ring.

Q30 / B competing elimination

Answer: (B) But-2-ene.

Why: Removal of a beta hydrogen from the more substituted neighbouring carbon forms the internal, more substituted double bond. Under ordinary small-base elimination conditions this alkene is favoured over terminal but-1-ene. The carbon skeleton does not rearrange in the simple E2 picture.

Why not the others: (A) is less substituted; (C) is substitution/reduction rather than elimination; (D) changes the carbon connectivity.

Key point: Compare alkene substitution from available beta positions.

Mathematics

Q31 / C hyperbola asymptote

Answer: (C) Slopes ±2/3.

Given: Original a=3,b=4; new a=6,b=4.

Formula/Principle: For $x^2/a^2 - y^2/b^2 = 1$, asymptotes $y = \pm(b/a)x$.

Step 1: Replace a by 2·3=6.

Step 2: b/a=4/6=2/3.

Worked derivation:

$$a' = 6,\ b' = 4 \quad \Rightarrow \quad y = \pm\frac{b'}{a'}x = \pm\frac{2}{3}x.$$

Why: Stretching the transverse semiaxis flattens both asymptotes, so the new slope magnitude must be below the original 4/3. The vertical semiaxis is explicitly held fixed, preventing both parameters from scaling together.

Why not the others: (A) retains original slopes; (B) incorrectly increases steepness; (D) inverts the changed semiaxis ratio.

Key point: Recompute the b/a ratio after a single-axis perturbation.

Q32 / D monotonicity interval

Answer: (D) The two outer intervals.

Given: Differentiable cubic.

Formula/Principle: f increases where f'(x)>0.

Step 1: f'=3x²−12x+9=3(x−1)(x−3).

Step 2: Product positive for x<1 or x>3, negative between.

Worked derivation:

$$f'(x) = 3(x-1)(x-3) > 0 \iff x < 1 \text{ or } x > 3.$$

Why: The derivative changes sign at each simple critical root. Testing a point in each interval avoids an incorrect blanket claim based only on the leading cubic coefficient; the endpoints are stationary rather than strictly positive-derivative points.

Why not the others: (A) is decreasing; (B) ignores the middle negative interval; (C) includes part of that decreasing interval.

Key point: Make a derivative sign chart, not just solve f'=0.

Q33 / A substitution and bounds

Answer: (A) (2√2−1)/3.

Given: A definite integral with a composite radical and matching derivative x dx.

Formula/Principle: Put u=1+x², so du=2x dx and the bounds x=0,1 become u=1,2.

Step 1: The integral becomes (1/2)∫₁²u^(1/2) du.

Worked derivation:

$$\int_0^1 x\sqrt{1+x^2}\,dx = \frac{1}{2}\int_1^2 u^{1/2}du = \frac{1}{3}[u^{3/2}]_1^2 = \frac{2\sqrt{2}-1}{3}.$$

Why: Changing both integrand and bounds avoids returning to x or applying an inconsistent upper limit. The x factor supplies the derivative required for substitution, and the transformed interval keeps the evaluation entirely in u.

Why not the others: (B) adds instead of subtracting the lower endpoint; (C) misses division by three; (D) drops a factor two at the upper bound.

Key point: Transform the limits along with the variable.

Q34 / B Minimum equivalence completion with a class-count constraint

Answer: (B) Six ordered pairs.

Given: R is already reflexive and symmetric; the chains force 1,2,3 into one class and 4,5 into another.

Formula/Principle: An equivalence class of size s contains all s^2 ordered pairs; merging classes of sizes a,b adds $2ab$ cross-pairs.

Step 1: Complete transitivity in $\{1,2,3\}$ by adding (1,3) and (3,1). The minimal closure then has classes of sizes 3,2,1.

Step 2: Exactly two classes requires merging at least two of these components; splitting a forced component is impossible.

Step 3: The possible additional merge costs are $2(3)(2) = 12$, $2(3)(1) = 6$ and $2(2)(1) = 4$.

Step 4: Merge $\{4, 5\}$ with $\{6\}$, giving total additions $2 + 4 = 6$.

Worked derivation:

$$N_{\min} = 2 + \min\{12, 6, 4\} = 6.$$

Why: Completing transitivity alone leaves three classes and therefore does not satisfy the extra class-count condition. Comparing merge costs proves that the chosen six additions are sufficient and minimal, not just one possible completion.

Why not the others: (A) stops at the three-class closure; (C) merges the size-three and size-one components; (D) merges the two largest components.

Key point: Establish forced classes, then optimise the permitted mergers.

Q35 / C matrix equation

Answer: (C) 0.

Given: A is invertible with determinant $5-6=-1$.

Formula/Principle: $X=A^{-1}B$, or solve each column of $AX=B$ independently.

Step 1: $A^{-1} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$.

Worked derivation:

$$X = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 5 & 1 \\ 13 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}, \qquad \operatorname{tr} X = 1 + (-1) = 0.$$

Why: Multiplying by B matters: $AX=B$ is not an inverse equation. Both columns must satisfy the corresponding columns of B, which the displayed X does; its diagonal entries have opposite signs.

Why not the others: (A) adds off-diagonal values with signs; (B) retains only the first diagonal; (D) ignores the negative second diagonal.

Key point: Solve the matrix equation before taking its trace.

Q36 / D binomial probability event

Answer: (D) 7/8.

Given: Three independent trials with P(tail)=1/2.

Formula/Principle: Use complement of no heads.

Step 1: P(all tails)=$(1/2)^3$=1/8.

Step 2: P(at least one head)=1−1/8=7/8.

Worked derivation:

$$P(H \geq 1) = 1 - P(TTT) = 1 - (1/2)^3 = 7/8.$$

Why: The event includes exactly one, two or three heads; using a single binomial term would omit valid outcomes. Complement counting handles all cases and is consistent with the certainty limit as the number of trials grows.

Why not the others: (A) is no heads; (B) is exactly two heads; (C) ignores one valid outcome.

Key point: Use the complement for an 'at least one' event.

Q37 / A inclusion-exclusion count

Answer: (A) 54.

Given: Divisibility counts floor(100/d).

Formula/Principle: Inclusion-exclusion for union, then subtract its multiples of five.

Step 1: Multiples of 2 or3: 50+33−16=67.

Step 2: Among multiples of5, also divisible by2 or3: 10+6−3=13.

Worked derivation:

$$N = (\lfloor 100/2 \rfloor + \lfloor 100/3 \rfloor - \lfloor 100/6 \rfloor) - (\lfloor 100/10 \rfloor + \lfloor 100/15 \rfloor - \lfloor 100/30 \rfloor) = 67 - 13 = 54.$$

Why: The union count double-counts multiples of six; the excluded five-multiples need their own union correction because multiples of 30 satisfy both exclusions. Subtracting all 20 multiples of five would discard some never counted initially.

Why not the others: (B) omits exclusion; (C) subtracts too many; (D) mishandles overlap.

Key point: Apply inclusion-exclusion to both inclusion and exclusion sets.

Q38 / B term independent of variable

Answer: (B) 84.

Given: General term chooses k factors of x^{-1}.

Formula/Principle: Exponent is 2(9−k)−k=18−3k; coefficient C(9,k).

Step 1: Set exponent zero, k=6.

Step 2: C(9,6)=C(9,3)=84.

Worked derivation:

$$T_{k+1} = \binom{9}{k} x^{18-3k}, \quad 18 - 3k = 0 \Rightarrow k = 6, \quad [x^0] = \binom{9}{6} = 84.$$

Why: The negative power shifts the exponent by three for each replacement, not by one. Identifying k first avoids selecting the central binomial coefficient merely because the expansion has nine factors.

Why not the others: (A) uses wrong k; (C) is a different binomial coefficient; (D) doubles a neighbouring term.

Key point: Solve the exponent condition before reading the coefficient.

Q39 / C series convergence under stated condition

Answer: (C) a=5/3.

Given: Common ratio r=a−1.

Formula/Principle: Geometric sum 1/(1−r) for |r|<1.

Step 1: 1/(2−a)=3 gives 2−a=1/3 and a=5/3.

Step 2: |5/3−1|=2/3<1, so solution obeys convergence condition.

Worked derivation:

$$S = \frac{1}{1-(a-1)} = \frac{1}{2-a} = 3 \Rightarrow a = \frac{5}{3}, \quad |a-1| = \frac{2}{3} < 1.$$

Why: Algebra alone can propose a parameter where the infinite-sum formula is invalid; checking the convergence interval is a separate required step. At a=2 the ratio is one and terms do not tend to zero.

Why not the others: (A) produces another sum; (B) produces 3/2; (D) diverges.

Key point: Validate the geometric condition after solving for a.

Q40 / D removable singularity

Answer: (D) a=3 and derivative 1.

Given: For x≠a, factor numerator to obtain f(x)=x+a.

Formula/Principle: Continuity requires b=lim f(x)=2a; derivative of extension x+a is one.

Step 1: 6=2a gives a=3.

Step 2: With the hole filled at x=3, f(x)=x+3 everywhere and f′(3)=1.

Worked derivation:

$$\frac{x^2-a^2}{x-a}=x+a,\quad b=2a=6\Rightarrow a=3,\quad f'(a)=\lim_{h\to 0}\frac{(a+h+a)-2a}{h}=1.$$

Why: The quotient formula is undefined at x=a until the separate assigned value fills the hole. Continuity fixes that value, after which the derivative follows from the linear extension; differentiating the unreduced quotient at its missing point is invalid.

Why not the others: (A) treats b as a directly; (B) doubles derivative; (C) makes both mistakes.

Key point: Repair continuity before differentiating at a removable singularity.

Q41 / A first-order linear solution

Answer: (A) 7/6.

Given: The unknown function satisfies an initial-value linear ODE; a second integral is requested.

Formula/Principle: Use integrating factor x to solve first, then integrate y(x)/x over [1,2].

Step 1: $(xy)'=2x^2$ gives $xy=(2/3)x^3+C$; y(1)=1 gives C=1/3.

Step 2: Thus $y(x)/x=(2/3)x+1/(3x^2)$.

Worked derivation:

$$\int_1^2 \frac{y(x)}{x}\,dx=\left[\frac{x^2}{3}-\frac{1}{3x}\right]_1^2=\left(\frac{4}{3}-\frac{1}{6}\right)-\left(\frac{1}{3}-\frac{1}{3}\right)=\frac{7}{6}.$$

Why: The initial condition supplies the C/x term in y; it contributes a nonzero amount to the requested integral. Solving only for y(2) would answer a different question.

Why not the others: (B) is a point-value from a simpler variant; (C) is y(2), not the integral; (D) arises from omitting the inverse-square contribution.

Key point: Solve the ODE before evaluating the functional of its solution.

Q42 / B plane intersection

Answer: (B) (t,1,2−t).

Given: Subtract plane equations.

Formula/Principle: Solve linear constraints, then let one free coordinate be t.

Step 1: 2y=2 gives y=1.

Step 2: x+z=2; set x=t, z=2−t.

Worked derivation:

$$(x+y+z)-(x-y+z)=2\Rightarrow y=1,\quad x+z=2\Rightarrow (x,y,z)=(t,1,2-t).$$

Why: Two nonparallel planes in three dimensions leave one free parameter. Substitution of the candidate into both equations checks not just direction but the correct offset of the line.

Why not the others: (A) has wrong y; (C) wrong y and sum; (D) makes first plane total four.

Key point: Subtract constraints to identify fixed and free coordinates.

Q43 / C cross-product orientation

Answer: (C) (−1/√2,0,1/√2).

Given: Orientation fixed as a×b, not b×a.

Formula/Principle: Cross product then divide by its length.

Step 1: a×b=(−2,0,2), norm 2√2.

Step 2: Normalise to (−1/√2,0,1/√2).

Worked derivation:

$$\mathbf{a}\times\mathbf{b}=(-2,0,2),\quad \frac{\mathbf{a}\times\mathbf{b}}{|\mathbf{a}\times\mathbf{b}|}=(-1/\sqrt{2},0,1/\sqrt{2}).$$

Why: Swapping vector order reverses the cross product, so the positive-z condition distinguishes the requested orientation. A raw perpendicular vector is insufficient when the question asks unit length.

Why not the others: (A) reverses orientation; (B) is not perpendicular to a; (D) has norm √2 rather than one.

Key point: Check orientation, orthogonality and norm.

Q44 / D affine variance transform

Answer: (D) a=−3,b=8.

Given: Var(aX+b)=a² Var(X), E(Y)=aE(X)+b.

Formula/Principle: Solve spread first, choose sign, then centre.

Step 1: 36=4a² gives |a|=3; a<0 gives a=−3.

Step 2: −1=−3(3)+b gives b=8.

Worked derivation:

$$a^2=36/4=9,\ a<0\Rightarrow a=-3;\quad -1=3a+b\Rightarrow b=8.$$

Why: Variance cannot determine the sign of a because it squares deviations; the stated negative slope resolves that ambiguity. The mean condition then fixes the translation independently.

Why not the others: (A) chooses positive slope; (B) solves mean with wrong sign; (C) satisfies neither negative-slope requirement nor mean.

Key point: Resolve scale sign before using the mean.

Q45 / A trig equation endpoints

Answer: (A) 7.

Given: tan x is undefined where cos x=0; the interval includes both endpoints.

Formula/Principle: For cos x≠0, sin x/cos x=2sin x cos x.

Step 1: Either sin x=0 (0,π,2π), or cos²x=1/2 (four quarter-angle roots).

Worked derivation:

$$\sin x(1-2\cos^2 x)=0,\quad x\in\{0,\pi,2\pi,\pi/4,3\pi/4,5\pi/4,7\pi/4\};\quad N=7.$$

Why: Multiplying by cos x is valid only after excluding its zeros, where the original tangent is undefined. Both closed endpoints are valid distinct x values.

Why not the others: (B) omits endpoints or two quarter-angle roots; (C) counts only sin x=0 roots; (D) omits one valid endpoint.

Key point: Preserve the original domain during trigonometric factorisation.

Physics

Q46 / B relative motion with shared origin

Answer: (B) 6 s.

Given: Same initial position; A displacement 6t, B displacement (1/2)(2)t²=t².

Formula/Principle: Equal displacements at meeting.

Step 1: 6t=t² gives t(t−6)=0.

Step 2: Exclude the initial t=0 meeting; next is t=6 s.

Worked derivation:

$$x_A = 6t, \quad x_B = t^2, \quad x_A = x_B \Rightarrow t = 0 \text{ or } 6\,\text{s}.$$

Why: Equal velocities are not required at a crossing; B has accelerated enough to catch A after an initially slower start. The shared-origin root at release is not the 'next' meeting.

Why not the others: (A) equates B's velocity to A prematurely; (C) overshoots the crossing; (D) doubles the correct time.

Key point: Equate positions and exclude the initial root.

Q47 / C unit conversion trap

Answer: (C) 100 m.

Given: Convert endpoint speed 72 km h⁻¹=20 m s⁻¹.

Formula/Principle: Displacement is area under v–t graph.

Step 1: Mean speed for linear rise is 10 m s⁻¹.

Step 2: Multiply by 10 s to get 100 m.

Worked derivation:

$$v_{max} = 72/3.6 = 20\,\text{m,s}^{-1}, \quad s = \tfrac{1}{2}(10\,\text{s})(20\,\text{m,s}^{-1}) = 100\,\text{m}.$$

Why: The graph's vertical unit is km/h while time is seconds; multiplying raw numbers would yield inconsistent units. The triangular area is half the rectangle because speed starts at zero.

Why not the others: (A) halves twice; (B) uses the numeric 72 without conversion; (D) uses the full rectangle.

Key point: Convert graph axes to consistent units before taking area.

Q48 / D pressure in accelerating vessel

Answer: (D) 6000 Pa.

Given: Effective downward acceleration in vessel frame is g+a=12 m s⁻².

Formula/Principle: Hydrostatic gauge pressure p=ρg_eff h.

Step 1: Calculate g_eff=12.

Step 2: p=1000·12·0.50=6000 Pa.

Worked derivation:

$$p_{gauge} = \rho(g+a)h = 1000(10+2)(0.50) = 6000\,\text{Pa}.$$

Why: An upward accelerating vessel behaves as though gravity were stronger; the free-surface pressure is the zero-gauge reference. Using g−a would correspond to downward acceleration of the support.

Why not the others: (A) uses g−a; (B) ignores support acceleration; (C) applies a wrong depth or factor.

Key point: Determine effective gravity before hydrostatic calculation.

Q49 / A drilled-body inertia

Answer: (A) (5/8)mR².

Given: Removed mass M/4 and remaining mass m=3M/4.

Formula/Principle: Subtract the small disc inertia from original, then convert M to m.

Step 1: I=(1/2)MR²−(1/2)(M/4)(R/2)²=(15/32)MR².

Step 2: M=4m/3 gives I=(5/8)mR².

Worked derivation:

$$I = \tfrac{1}{2}MR^2 - \tfrac{1}{2}\tfrac{M}{4}\tfrac{R^2}{4} = \tfrac{15}{32}MR^2 = \tfrac{5}{8}mR^2.$$

Why: The removed area is one quarter of the original, but its moment has an additional radius-squared factor. The problem asks for the answer in terms of remaining mass, making (15/32)MR² an incomplete final form.

Why not the others: (B) treats remaining material as a solid disc; (C) overweights the rim; (D) leaves original mass M in the coefficient.

Key point: Subtract inertia and convert mass basis at the end.

Q50 / B engine-cycle energy bookkeeping

Answer: (B) 200 J and 40%.

Given: Net internal energy change over complete cycle is zero.

Formula/Principle: W=Q_in−Q_out; η=W/Q_in.

Step 1: W=500−300=200 J.

Step 2: η=200/500=0.40=40%.

Worked derivation:

$$\Delta U_{cycle} = 0, \quad W = 500 - 300 = 200\,\text{J}, \quad \eta = 200/500 = 0.40.$$

Why: A cyclic engine returns to its initial state, so heat input not rejected becomes work. Efficiency uses input heat as denominator, not rejected heat or total heat exchanged.

Why not the others: (A) treats rejected heat as work; (C) violates energy balance; (D) divides by the wrong quantity.

Key point: Close the cycle energy balance before calculating efficiency.

Q51 / C spring-combination period

Answer: (C) 2.

Given: Series equivalent k/2; parallel equivalent 2k.

Formula/Principle: T=2π√(m/k_eff).

Step 1: T_s=2π√(2m/k); T_p=2π√(m/(2k)).

Step 2: Ratio √[(2m/k)/(m/(2k))]=2.

Worked derivation:

$$\frac{T_s}{T_p} = \sqrt{\frac{k_p}{k_s}} = \sqrt{\frac{2k}{k/2}} = 2.$$

Why: Series coupling softens the system and lengthens its period; parallel coupling stiffens it and shortens period. The ratio cancels mass and the common 2π factor.

Why not the others: (A) reverses configurations; (B) assumes same stiffness; (D) uses stiffness ratio without its square root.

Key point: Combine spring constants before applying the period law.

Q52 / D interference with source phase

Answer: (D) Constructive.

Worked derivation:

$$\Delta\phi_{path} = 2\pi(\lambda/2)/\lambda = \pi, \quad \Delta\phi_{total} = \pi + \pi = 2\pi.$$

Why: Both source phase and travel phase affect arrival. A half-wavelength geometric delay alone would be destructive for in-phase sources, but the supplied initial π shift adds another π and restores alignment.

Why not the others: (A) ignores source phase; (B) contradicts coherence; (C) invents a frequency difference.

Key point: Add initial and path phase differences before classifying interference.

Q53 / A electric potential from field region

Answer: (A) (−E₀a,0).

Given: Signed field is piecewise constant; potential difference is negative signed field area.

Formula/Principle: $V(x) - V(0) = -\int_0^x E_x dx$.

Step 1: To 2a, area +2E₀a−E₀a=+E₀a, so potential difference −E₀a.

Step 2: To 3a, area +2E₀a−2E₀a=0.

Worked derivation:

$$\Delta V_{0\to 2a} = -[2E_0 a - E_0 a] = -E_0 a, \quad \Delta V_{0\to 3a} = -[2E_0 a - E_0(2a)] = 0.$$

Why: The negative field interval partially then fully cancels the earlier positive area. Reading only magnitudes or treating potential as proportional to the instantaneous field at an endpoint misses the accumulated integral.

Why not the others: (B) drops the minus sign; (C) cancels too early; (D) miscounts interval widths.

Key point: Integrate signed field before negating for potential.

Q54 / B internal resistance terminal voltage

Answer: (B) 2 A, 10 V.

Given: Total series resistance 5+1=6 Ω.

Formula/Principle: I=ε/(R+r), terminal V=ε−Ir=IR.

Step 1: I=12/6=2 A.

Step 2: V=12−2·1=10 V.

Worked derivation:

$$I = \frac{12}{5+1} = 2\,\text{A}, \quad V_{term} = 12 - (2)(1) = 10\,\text{V}.$$

Why: Internal resistance consumes part of emf under load, so terminal voltage is below open-circuit 12 V. The external resistor check IR=2·5=10 V confirms the result.

Why not the others: (A) ignores internal drop; (C) neglects r in current; (D) confuses voltage and current.

Key point: Distinguish emf from loaded terminal voltage.

Q55 / C multi-stage decay bookkeeping

Answer: (C) 100 nuclei.

Given: Two half-lives first, then one more after removing daughter.

Formula/Principle: Parent halves every 2 h; daughter formed equals parent loss.

Step 1: A after 4 h=800/4=200; old B removed.

Step 2: One further half-life takes A to100; new B=200−100=100.

Worked derivation:

$$N_A(4h) = 800(1/2)^2 = 200, \quad N_A(6h) = 100, \quad \Delta N_B = 200 - 100 = 100.$$

Why: Removing stable daughter does not reset the surviving radioactive parent population. Only the loss during the last stage counts as newly formed B; adding the earlier 600 removed B would answer a different question.

Why not the others: (A) takes an extra half-life; (B) counts all remaining A as decayed; (D) uses the first-stage production.

Key point: Track surviving parent across an intervention.

Q56 / D inductor energy from current

Answer: (D) 3 J.

Given: Initial and final current through same L.

Formula/Principle: U=(1/2)LI².

Step 1: U_i=0.25·4=1 J.

Step 2: U_f=0.25·16=4 J; increase 3 J.

Worked derivation:

$$\Delta U = \tfrac{1}{2}(0.50)(4^2 - 2^2) = 0.25(12) = 3\,\text{J}.$$

Why: Stored energy scales with current squared, so doubling current quadruples total energy rather than doubling it. The question asks extra energy, requiring subtraction of the initial one joule.

Why not the others: (A) reports initial energy; (B) scales linearly; (C) reports final energy.

Key point: Subtract initial from final quadratic energy.

Q57 / A prism deviation

Answer: (A) 60°.

Given: Symmetric path at minimum deviation.

Formula/Principle: $n = \sin[(A+\delta_m)/2]/\sin(A/2)$.

Step 1: sin[(60+δ)/2]=√3·sin30=√3/2=sin60.

Step 2: Physical angle gives (60+δ)/2=60, so δ=60°.

Worked derivation:

$$\sqrt{3} = \frac{\sin((60^\circ + \delta_m)/2)}{\sin 30^\circ} \Rightarrow \delta_m = 60^\circ.$$

Why: The minimum-deviation ray has equal internal refraction angles, which underlies the given relation. Choosing the supplementary sine solution would violate the geometry of this prism path.

Why not the others: (B) makes the sine too small; (C) uses a different incident angle; (D) overstates the bend.

Key point: Apply symmetric minimum-deviation geometry.

Q58 / B phasor phase and impedance

Answer: (B) 1 A, lagging.

Given: Voltage triangle components 30 and +40 V, hypotenuse50 V.

Formula/Principle: $V_R = IR$; positive net inductive component means current lags voltage.

Step 1: I=30 V/30 Ω=1 A.

Step 2: $\tan\phi = 40/30 = 4/3$, with inductive lag.

Worked derivation:

$$I = V_R/R = 30/30 = 1\,\text{A}, \quad \tan\phi = V_X/V_R = 4/3, \quad \phi > 0 \text{ (current lags)}.$$

Why: Phasor components add at right angles to make 50 V; dividing the total 50 V by R alone would ignore reactance. A positive upward inductive component sets the lag direction, not just the phase magnitude.

Why not the others: (A) divides total V by R and reverses phase; (C) gets current but reverses phase; (D) still ignores reactive impedance.

Key point: Read both magnitude and sign from the phasor triangle.

Q59 / C rectifier ripple comparison

Answer: (C) 50 Hz then 100 Hz.

Worked derivation:

$$f_{half} = f_{in} = 50\,\text{Hz}, \qquad f_{full} = 2f_{in} = 100\,\text{Hz}.$$

Why: Half-wave passes one current pulse per input period; full-wave flips the negative half-cycle to make two positive pulses per period. This is ripple repetition frequency before smoothing, not the frequency of the original alternating voltage.

Why not the others: (A) reverses pulse counts; (B) ignores the second full-wave pulse; (D) doubles half-wave too.

Key point: Count output pulses per input cycle.

Q60 / D thin-film phase reversal

Answer: (D) 100 nm.

Given: Air-to-film reflection gains π phase; film-to-lower-index glass reflection gains none.

Formula/Principle: With one phase reversal, constructive reflection requires $2nt = (m + 1/2)\lambda$.

Step 1: For least positive t use m=0.

Step 2: $t = \lambda/(4n) = 600/(4{\cdot}1.5) = 100$ nm.

Worked derivation:

$$2nt = \tfrac{1}{2}\lambda \Rightarrow t_{min} = \frac{600\,\text{nm}}{4(1.5)} = 100\,\text{nm}.$$

Why: The two reflected rays differ by one interface phase reversal; the optical round trip must add another π to align them. If the substrate index exceeded the film index, both reflections would reverse and the thickness rule would change.

Why not the others: (A) halves the optical path twice; (B) uses the two-reversal condition; (C) miscounts n.

Key point: Count reflection phase reversals before path difference.
